MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6386

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, remote-cutoff

Outline	---	6-1 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	h	2k	2g	2a	sd	1a	1g	1k	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk/k	Rg/g	Ik/k	Pp/a	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	ft
Maximum:	6.9	330	0	100	---	0.5	20	1.65	+165	Note 1
Minimum:	5.7	---	-50	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	0	200	---	---	---	---	---

GENERAL:

Qualification - Not Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	330	370	mA
1336	Heater-cathode leakage	2		0.4	II	Ihk	---	10	µAdc
1266	Total grid current	2,3	Rg/g = 0.5 Meg	0.4	II	Ic	0	-0.3	µAdc
1256	Electrode current (anode)	2,3		0.4	II	Ib	6.6	12.6	mAdc
1306	Transconductance (1)	2		0.4	II	Sm	3,200	4,800	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	2		2.5	I	R	---	100	Meg
1256	Electrode current (anode), difference between sections			2.5	I	Ib	---	2.5	mAdc
1306	Transconductance (2)	2	Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1306	Transconductance (3)	2	Ec = -22 Vdc	2.5	I	Sm	---	100	µmhos
1266	Grid emission	2,5	Ef = 7.5 V; Rg/g = 0.5 Meg; Ec = -30 Vdc	2.5	I	Isc	0	-0.5	µAdc
2201	Noise and microphonics	4	Ef = 6.3 Vac; Ehk = 0; Ebb = 250 Vdc; Ecal = 100 mVac; Rp = 10,000 ohms; Rk = 1,000 ohms	2.5	I	---	---	---	---
1316	Amplification factor	2		---	---	Mu	12	22	---
1331	Direct-interelectrode capacitance	2	No shield	6.5	Code E	Cgp	---	1.4	pF
		2				Cin	1.5	2.5	pF
		2				Cout	0.8	1.4	pF
						Cgg	---	0.006	pF
						Cpp	---	0.12	pF
		2				Chk	2.8	4.4	pF
---	Cascade plate current	6		---	---	Ib	6.5	14.5	mAdc
---	Cascade transconductance	6		---	---	Sm	2,600	5,400	µmhos
---	Distortion	7		---	---	---	---	1.2	%

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1031	High-frequency vibration	4	Ec = -2 Vdc; Rp = 2,000 ohms; Rk/k = 0	---	---	Ep	---	130	mVac
1041	Shock	8	Accel = 450 G; Ehk = +100 Vdc	6.5	Note 9	---	---	---	---
1031	Vibration-fatigue			6.5	Note 9	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration	4	Ec = -2 Vdc; Rp = 2,000 ohms; Rk/k = 0	---	---	Ep	---	200	mVac
1336	Heater-cathode leakage	2		---	---	Ihk	---	30	µAdc
1306	Transconductance (1)	2		---	---	Sm	2,800	---	µmho
1266	Total grid current	2	Rg/g = 0.5 Meg	---	---	Ic	0	-0.6	µAdc
1121	Base Strain			---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = +135 Vdc; Rk = 0. Ec = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage	2		Ihk	---	20	µAdc
1516	Stability life		Eb = 150 Vdc; Ehk = +135 Vdc; Rg/g = 0.5 Meg; TA = room; Rk/k = 470 ohms	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes	2		ΔSm t	---	10	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
1501	Intermittent life	10,11	Eb = 150 Vdc; Ehk = +135 Vdc; Rg/g = 0.5 Meg; Rk/k = 470 ohms; TA = room; TE = +165°C (min)	---	---	---	---
---	Intermittent life-test end points (500 hours):	12					
1266	Total grid current	2	Rg/g = 0.5 Meg	Ic	0	-0.3	µAdc
1301	Heater current			If	330	375	mA
1306	Change in transconductance (1) of individual tubes	2		ΔSm t	---	20	%
1306	Transconductance (2)	2	Ef = 5.7 V	ΔSm Ef	---	15	%
1336	Heater-cathode leakage	2		Ihk	---	10	µAdc
1211	Insulation of electrodes	2		R	60	---	Meg
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%
---	Intermittent life-test end points (1,000 hours):	12					
1266	Total grid current	2	Rg/g = 0.5 Meg	Ic	0	-0.3	µAdc
1301	Heater current			If	330	380	mA
1306	Change in transconductance (1) of individual tubes	2		ΔSm t	---	25	%
1306	Transconductance (2)	2	Ef = 5.7 V	ΔSm Ef	---	20	%
1336	Heater-cathode leakage	2		Ihk	---	10	µAdc
1211	Insulation of electrodes	2		R	50	---	Meg
1511	Cathode interface life		Ef = 6.9 V	Ri	---	50	Ohms

NOTES:

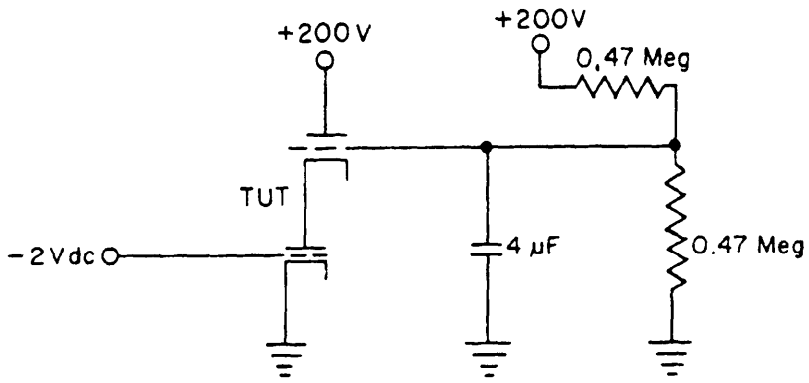
1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Test each unit separately.
3. This test shall be performed at the conclusion of the holding period.
4. Tie 1k to 2k; 1g to 2g; 1a to 2a.
5. Prior to this test, the TUT shall be preheated a minimum of 5 minutes with all sections operating at the following conditions. The 3-minute test shall not be permitted. Test at the conditions specified within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ec Vdc	Eb Vdc	Rk/k Ohms	Rg/g Meg
7.5	0	150	470	0.5

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NOTES: - Continued

6. Read in cascade circuit shown on figure 1.



Read the TUT both ways (sections interchanged).

FIGURE 1. Cascade circuit.

7. The TUT shall be tested as shown on figure 2. A 0.7 V rms 1,000 Hz signal shall be applied at point A. The 1,000 Hz output and the 3,000 Hz output shall be measured at point B. The ratio of the 3,000 Hz output to the 1,000 Hz output shall be calculated and shall be converted to percent. The TUT shall be read both ways (sections interchanged).

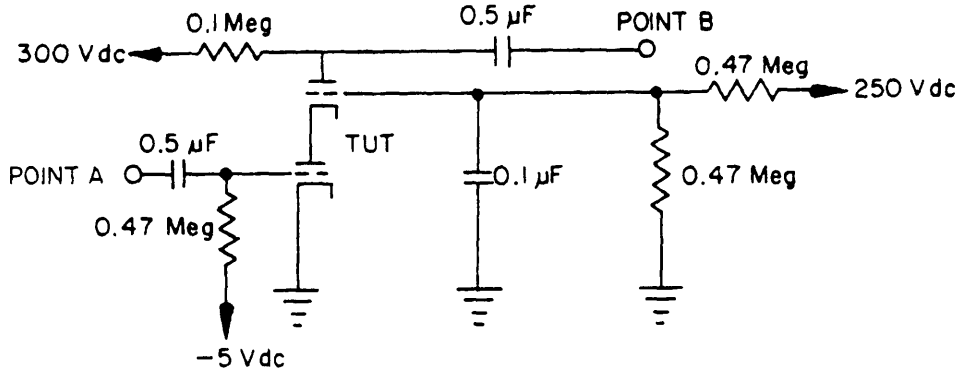


FIGURE 2. Distortion circuit.

8. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
9. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
10. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
11. The life-test sample shall consist of 20 tubes, and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

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NOTES: - Continued

12. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples. Total defectives shall be 3 and 6, respectively, for 500 hours, and 4 and 8, respectively, for 1,000 hours.
13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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